Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/20/2004
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Project Title:	Mechanics of Granular Materials		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Gravitational		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-0428	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	12/01/1992	**End Date:**	10/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	16
No. of PhD Candidates:	3	**No. of Master' Degrees:**	11
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Al Shibli, K. (Louisiana State University)		
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Ground-based displacement-controlled triaxial experiments are conducted on a cohesionless granular material at the lowest effective confining pressures possible, that do not result in material instability, to assess constitutive properties, stability phenomena, and control parameters that will be applied to in-space experiments on 75 mm (diameter) and 150 mm (length) right cylindrical specimens. The ground-based tests on similar-sized specimens are conducted in the range 3.5-69 kPa, while the microgravity tests will be conducted at effective confinement levels in the range 0.05-1.30 kPa. The displacement-controlled mode of loading confined specimens was chosen mainly to maintain overall specimen-apparatus stability while strain-softening resulting from continuous or discontinuous bifurcation and discontinuous deformation fields are allowed to take place. Optical and other noncontacting displacement-sensing techniques are used to measure specimen response during experimentation. Prescribed displacements are transmitted in terms of loading, unloading, and reloading historics, while volume change is measured in "drained" tests, and pore fluid pressure is measured in "undrained" isochoric tests. Confinement pressure is transmitted to the granular material

Task Description:	assembly through a thin flexible latex membrane surrounding the specimen. A subangular and uniform Ottawa quartz sand constitutes the specimen. Specimens tested both in space and on ground have been subjected to nondestructive and destructive (thin-slicing) testing to assess degrees of material uniformity and isotropy before and after experimentation. It appears that instability phenomenon associated with specimens of certain configurations result in curved internal surfaces of localized deformation and high rates of dilatancy, whose structure depends on bifurcation mode. Results from in-space experiments have shown that the friction angles at very low stresses can be as high as 75 degrees (range 75 to 56 degrees, with decreasing angles at higher stresses), while typical friction angles observed in terrestrial experiments at higher confinement levels are in the range of 42 to 40 degrees. Also, it was observed that the dilatancy angle, which is a measure of volumetric change and expansion was as high as 30 degrees, while typical terrestrial experiments show dilatancy angles in the range of 10 degrees. It was also observed that the elastic moduli were substantially higher (20 MPa) than was predicted for granular materials at very low confinement stress levels. It has also been observed that loose to medium dense granular assemblies tend to dilate rather than contract in volume at very low stress levels. Computed Tomography (Gamma-ray and X-ray) have shown extensive interior localized deformation or slip patterns in the specimens, which have not before been observed. The slip patterns, which clearly show local areas of localized volume expansion (increased porosity) appear in a very regular and periodic manner, and display turbine-fan like configurations with periodic spacing, which appear to be function of grain size, specimen shape, effective confining pressure and various moduli and material constants (constitutive properties). The curious observation is that these patterns become visible long after global material instability has occurred, and at the stage when the overall deformations have almost reached the residual strength or critical-state level. These findings are very different from what has earlier been observed in plane strain experiments, which show single interior surfaces, and coinciding instability and appearance of localized deformations. These findings have great importance in engineering mechanics and engineering disciplines, especially earthquake engineering. The objective of this research is to gain a quantitative and qualitative understanding of the mechanical behavior of cohesionless granular materials, which are either dry or water saturated, under very low effective confining pressures subjected to either drained or undrained conditions under static as well as cyclic conditions. The objective is also to understand how the very high friction and dilatancy angles arise, what the underlying mechanisms are, and how volume-change phenomena which are both counter-intuitive and in contrast to thermodynamics concepts can occur. The aim is also to understand how the water-saturated granular structure with relatively high strength, stability and stiffnes stransitions to a liquid mixture or composite with near zero strength and shear stiffness. The research has shown that the quasi-static deformation and strength behavior of granular materials at low stress levels is very different from conventional concepts and what current models predict. The findings will have impact on structure-medium analysis techniques and foundation design, especially in earthquake prone areas. The findings are also important in understanding phenomena related to strength and stiffness investigations of lunar and planetary soils (regolith).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The findings will help understand strength, stability and stiffness behavior of both dry and water-saturated granular soils subjected to static as well as dynamic (earthquake induced strong-motion) loading and deformations at low effective stress levels, which may influence the way foundations, earth structures, soil-structure systems, etc. are designed and constructed to enhance safety and economic designs.
Task Progress:	During the past fiscal year the investigator team has made excellent progress in both areas of experimental and analytical modeling of the constitutive and stability behavior of granular materials, especially under low confinement stress levels. The internal structure and fabric of the specimens have been carefully studied by means of Computer Tomography, and we have made discoveries regarding formation and progress of localized deformations in terms of multiple shear bands and associated volume change. We have also gained better understanding how instability and deformation initiate and develop in the apparent absence of formation of shear bands, which only seem to develop at a later stage. We have also developed analytical and computational techniques for simulating strength, deformation and volume change behavior of both small-size and large regions of granular soils or regolith. We are still developing analysis and conceptual understanding for the observed high-modulus properties at low stress levels. We are also making progress in using the video- and photographic records obtained during STS-107 to understand volume change behavior and material instability phenomena. We are also making progress in understanding solid-fluid transition phenomena and associated instabilities.
Bibliography Type:	Description: (Last Updated: 03/14/2017)
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